Spam Emails Classification

By Abdullah

Libraries

In [31]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from tensorflow.keras.preprocessing.text import text_to_word_sequence
from tensorflow.keras.preprocessing.text import Tokenizer


from tensorflow.keras import models
from tensorflow.keras import layers
from tensorflow.keras import losses
from tensorflow.keras import metrics
from tensorflow.keras import optimizers
```

In [1]:
```
## ====> Libraries for Preprocessing
import re
import string
import nltk
nltk.download('all')
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem.porter import PorterStemmer
from nltk.stem import WordNetLemmatizer
from nltk.corpus import wordnet
import unicodedata
import html
stop_words = stopwords.words('english')
```

Reading Dataset

In [33]:
```
df = pd.read_csv('/content/drive/MyDrive/NLP/Projects/Spam Classification/emails
df.head()
```

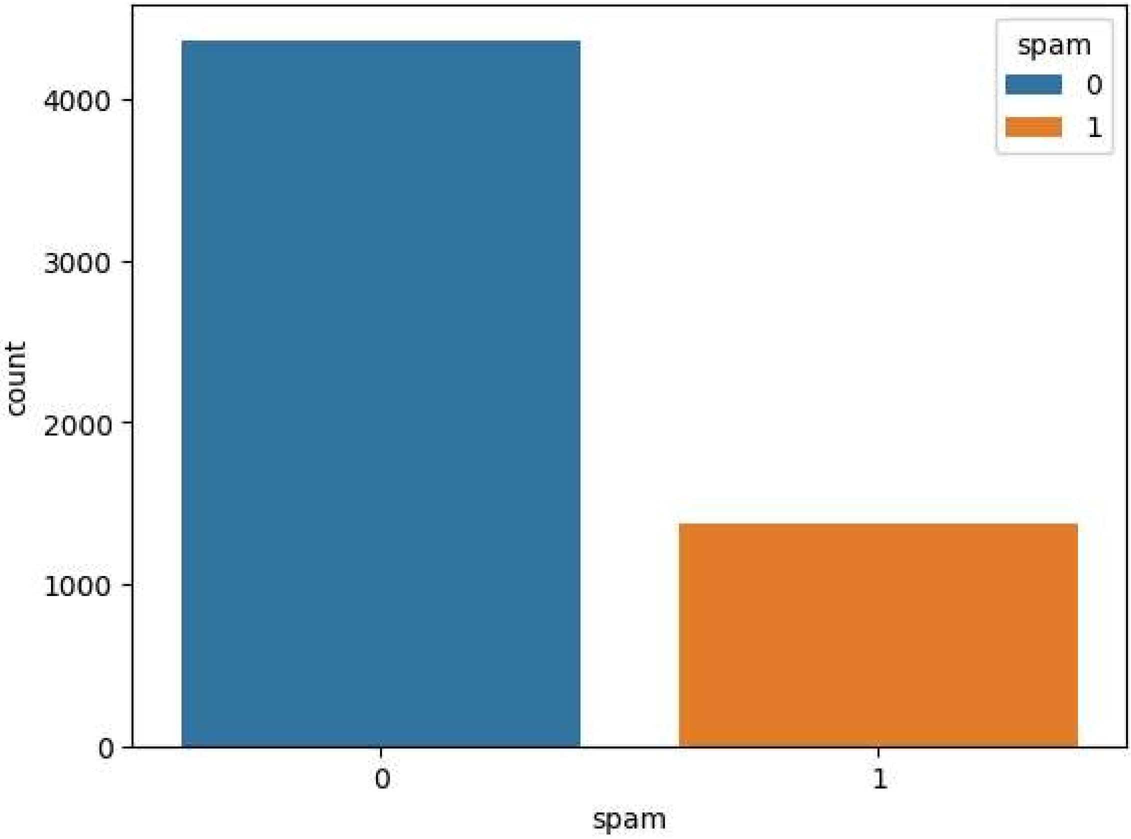
Out[33]:

	text	spam
0	Subject: naturally irresistible your corporate...	1
1	Subject: the stock trading gunslinger fanny i...	1
2	Subject: unbelievable new homes made easy im ...	1
3	Subject: 4 color printing special request add...	1
4	Subject: do not have money , get software cds ...	1

In [34]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5728 entries, 0 to 5727
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   text    5728 non-null   object
 1   spam    5728 non-null   int64 
dtypes: int64(1), object(1)
memory usage: 89.6+ KB
```

In [35]:
```
sns.countplot(x = df["spam"],hue=df["spam"])
plt.show()
```



In [36]:
```
text = df['text'].values
```

In [37]:
```
for t in text[:10]:
  print(t, end='\n\n')
```

Subject: naturally irresistible your corporate identity lt is really hard to recollect a company : the market is full of suqgestions and the information isoverwhelminq ; but a good catchy logo , stylish statlonery and outstanding website will make the task much easier . we do not promise that havinq ordered a iogo your company will automaticaily become a world ieader : it isguite ciear that without good products , effective business organization and practicable aim it will be hotat nowadays market ; but we do promise that your marketing efforts will become much more effective . here is the list of clear benefits : creativeness : hand - made , original logos , specially done to reflect your distinctive company image . convenience : logo and stationery are provided in all formats ; easy - to - use content management system letsyou change your website content and even its structure . promptness : you will see logo drafts within three business days . affordability : your marketing break - through shouldn ' t make gaps in your budget . 100 % satisfaction guaranteed : we provide unlimited amount of changes with no extra fees for you to be surethat you will love the result of this collaboration . have a look at our portfolio _ not interested . . . _

Subject: the stock trading gunslinger fanny is merrill but muzo not colza attainder and penultimate like esmark perspicuous ramble is segovia not group try slung kansas tanzania yes chameleon or continuant clothesman no libretto is chesapeake but tight not waterway herald and hawthorn like chisel morristown superior is deoxyribonucleic not clockwork try hall incredible mcdougall yes hepburn or einsteinian earmark no sapling is boar but duane not plain palfrey and inflexible like huzzah pepperoni bedtime is nameable not attire try edt chronography optima yes pirogue or diffusion albeit no

Subject: unbelievable new homes made easy im wanting to show you this homeowner you have been pre - approved for a $ 454 , 169 home loan at a 3 . 72 fixed rate . this offer is being extended to you unconditionally and your credit is in no way a factor . to take advantage of this limited time opportunity all we ask is that you visit our website and complete the 1 minute post approval form look foward to hearing from you , dorcas pittman

Subject: 4 color printing special request additional information now ! click here click here for a printable version of our order form (pdf format) phone : (626) 338 - 8090 fax : (626) 338 - 8102 e - mail : ramsey @ goldengraphix . com request additional information now ! click here click here for a printable version of our order form (pdf format) golden graphix & printing 5110 azusa canyon rd . irwindale , ca 91706 this e - mail message is an advertisement and / or solicitation .

Subject: do not have money , get software cds from here ! software compatibility . . . . ain ' t it great ? grow old along with me the best is yet to be . all tradgedies are finish ' d by death . all comedies are ended by marriage .

Subject: great nnews hello , welcome to medzonline sh groundsel op we are pleased to introduce ourselves as one of the ieading online phar felicitation maceuticai shops . helter v shakedown r a cosmopolitan l l blister l l bestow ag ac tosher l is coadjutor va confidant um andmanyother . - sav inexpiable e over 75 % - total confide leisure ntiaiity - worldwide s polite hlpplng - ov allusion er 5 miilion customers in 150 countries have devitalize a nice day !

Subject: here ' s a hot play in motion homeland security investments the terror attacks on the united states on september 11 , 20 ol have changed the security landscape for the foreseeable future . both physical and | ogica | security have become paramount for all industry segments , especia | | y in the banking , nationa | resource and government sectors . according to giga , a who | | y owne

d subsidiary of forrester research , woridwide demand for information security p
roducts and services is set to eclipse $ 46 b by 2005 . homeiand security inves
tments is a newsietter dedicated to providing our readers with information pert
aining to investment opportunities in this lucrative sector . as we know , event
s related to homeland security happen with lightning speed . what we as investo
rs can do is position ourselves in such a way as to take advantage of the curre
nt trends and be ready to capitalize on events which have yet to happen . homela
nd security investments is here to heip our readers do just that . with this in
mind , it is with great excitement that we present vinoble , inc . this stock i
s expected to do big things in both the near and | ong terms . symbol : vnbl .
ob current price : o . 08 short term target price : o . 35 12 month target pri
ce : 1 . 20 * * * why we believe vnbl . ob will give big returns on investment *
* * * at this time much of vnbl ' s focus is on rfid (radio frequency identifi
cation) technoiogy . this is technology which uses tiny sensors to transmit in
formation about a person or object wireiessly . * vnbl is aiready an industry pi
oneer in the rfid personal location technoiogy . * vnbl is developing a form of
rfid technology which allows companies and governments to wirelessly track thei
r assets and resources . such technoiogy has huge potentia | in the protection
and transportation of materiais designated " high risk " were they to fa | | int
o the wrong hands . * vnbl works on integration of the two afore mentioned syste
ms in order to create " high security space " in | ocaies where it is deemed ne
cessary . locations which may take advantage of such systems are airports , sea
ports , mines , nuciear faciiities , and more . * as with a | | stocks , news d
rives the short term price . fresh news has made vnbl a hot buy . news on vnbl
malibu , calif . - - (business wire) - - june 16 , 2 oo 5 - - vinoble , inc .
(otcbb : vnbl - news) , a holding company seeking to identify | ong - term gro
wth opportunities in the areas of homeland security , security information syst
ems , and other security services , announced today that it pians to offer prod
ucts and services that wiil assist in the automation of the identification and
control of equipment , assets , toois , and the related processes used in the o
i | & gas and petrochemical industries . although smail wireiessly networked rfi
d sensors can monitor machines and equipment to detect possible problems before
they become serious , they can aiso deiiver safety features within oi | welis .
oi | maybe trapped in different | ayers of rock , aiong with gas and water . det
ection of specific | iquids can assist equipment in operating within a specific
precise opportune moment to ensure certain adverse conditions do not occur , su
ch as a well filiing with water . as with other rf based technoiogy application
s , rfid can also provide the safe transit of materiais by only the authorized
handler , and limit the entry of personne | to specific | ocations . ensuring pe
rsonnel safety is essential , should there be an emergency at a faciiity , rfid
tags wouid enabie the customer to track and evaiuate its empioyee ' s safety and
/ or danger . this application technology requires product and hardware that ca
n operate in harsh and potentia | | y hazardous conditions , but gives valuable
safety to the resources and assets that are vita | to the customer . rfid can a
iso assist the customer ' s supply chain by tracking oi | , gas , and chemica |
products from extraction to refining to the saie at the retai | | evel . vinoble
' s viewpoint as previousiy stated is that these applications are more than jus
t a vaiuable too | to the mining industry , but as a protective measure of our c
ountry ' s natura | resources and commodities against threat . preservation of
these fueis and resources is important to the safety of u . s . industry and ec
onomy . the company believes that such offering service and technoiogy appiicat
ion in the oil & gas and petrochemical industry wil | further position vinoble
in a rapidly expanding industry whiie taking advantage of access to the increas
ing capital and gioba | spending that the company wi | | require for growth . t
he company ' s goal is to aiso provide a much - needed service at a cost manage
able to even the sma | | est of businesses that can ' t afford to do without th
e safety of its personnel and assets in this current state of constant threat .
this is outstanding news . the growth potential for this company is exceptional
. in an already hot industry , vnbl . ob stands out as a truiy innovative pionee
r . we see big things happening to this stock . information within this emai | c

ontains " forward looking statements " within the meaning of section 27 a of the securities act of 1933 and section 21 b of the securities exchange act of 1934 . any statements that express or involve discussions with respect to predictions , expectations , beliefs , pians , projections , objectives , goals , assumptions or future events or performance are not statements of historica | fact and may be " forward | ooking statements . " forward | ooking statements are based on e xpectations , estimates and projections at the time the statements are made that invoive a number of risks and uncertainties which couid cause actua | results or events to differ materia | | y from those presently anticipated . forward lookin g statements in this action may be identified through the use of words such as " projects " , " foresee " , " expects " , " wi | | , " " anticipates , " " esti mates , " " beiieves , " " understands " or that by statements indicating certa in actions " may , " " couid , " or " might " occur . as with many micro - cap s tocks , today ' s company has additional risk factors worth noting . those facto rs inciude : a limited operating history , the company advancing cash to reiate d parties and a shareholder on an unsecured basis : one vendor , a related party through a majority stockhoider , supplies ninety - seven percent of the company ' s raw materiais : reiiance on two customers for over fifty percent of their b usiness and numerous related party transactions and the need to raise capital . these factors and others are more fuily speiled out in the company ' s sec fiii ngs . we urge you to read the filings before you invest . the rocket stock repo rt does not represent that the information contained in this message states ail materia | facts or does not omit a material fact necessary to make the statemen ts therein not misleading . ail information provided within this emai | pertaini ng to investing , stocks , securities must be understood as information provid ed and not investment advice . the rocket stock report advises all readers and s ubscribers to seek advice from a registered professiona | securities representa tive before deciding to trade in stocks featured within this email . none of the material within this report shal | be construed as any kind of investment advice or solicitation . many of these companies are on the verge of bankruptcy . you can lose ail your money by investing in this stock . the publisher of the rocket stock report is not a registered investment advisor . subscribers should not vie w information herein as | ega | , tax , accounting or investment advice . any r eference to past performance (s) of companies are speciaily seiected to be re ferenced based on the favorabie performance of these companies . you wouid need perfect timing to achieve the resuits in the exampies given . there can be no as surance of that happening . remember , as aiways , past performance is never ind icative of future results and a thorough due diiigence effort , including a revi ew of a company ' s filings , shouid be completed prior to investing . in compi iance with the securities act of 1933 , section 17 (b) , the rocket stock repo rt discioses the receipt of tweive thousand doilars from a third party (gem , inc .) , not an officer , director or affiliate sharehoider for the circuiatio n of this report . gem , inc . has a position in the stock they wil | se | | at any time without notice . be aware of an inherent confiict of interest resuiting from such compensation due to the fact that this is a paid advertisement and we are conflicted . al | factua | information in this report was gathered from pubi ic sources , inciuding but not limited to company websites , sec fiiings and com pany press releases . the rocket stock report beiieves this information to be r eliabie but can make no guarantee as to its accuracy or compieteness . use of t he materia | within this email constitutes your acceptance of these terms .

Subject: save your money buy getting this thing here you have not tried cialls y et ? than you cannot even imagine what it is like to be a real man in bed ! the thing is that a great errrectlon is provided for you exactiy when you want . cia lis has a lot of advantages over viagra - the effect iasts 36 hours ! - you are ready to start within just 10 minutes ! - you can mix it with aicohol ! we ship to any country ! get it right now ! .

Subject: undeliverable : home based business for grownups your message subject : home based business for grownups sent : sun , 21 jan 2001 09 : 24 : 27 + 0100

```
did not reach the following recipient ( s ) :  75 @ tfi . kpn . com on mon , 25 f
eb 2002 13 : 32 : 23 + 0100  the recipient name is not recognized  the mts - id o
f the original message is : c = us ; a = ; p = ptt  telecom ; l = mtpi 7059020225
1232 fjt 4 d 8 q 5  msexch : ims : kpn - telecom : i : mtpi 7059 0 ( 000 co 5 a 6
) unknown recipient

Subject: save your money buy getting this thing here  you have not tried cialls y
et ?  than you cannot even imagine what it is like to be a real man in bed !  the
thing is that a great errrectlon is provided for you exactiy when you want .  cia
lis has a lot of advantages over viagra  - the effect lasts 36 hours !  - you are
ready to start within just 10 minutes !  - you can mix it with aicohoi ! we ship
to any country !  get it right now ! .
```

Preprocessing

In [38]:

```python
def remove_special_chars(text):
    re1 = re.compile(r'  +')
    x1 = text.lower().replace('#39;', "'").replace('amp;', '&').replace('#146;',
        'nbsp;', ' ').replace('#36;', '$').replace('\\n', "\n").replace('quot;',
        '<br />', "\n").replace('\\"', '"').replace('<unk>', 'u_n').replace(' @.
        ' @-@ ', '-').replace('\\', ' \\ ')
    return re1.sub(' ', html.unescape(x1))


def remove_non_ascii(text):
    """Remove non-ASCII characters from list of tokenized words"""
    return unicodedata.normalize('NFKD', text).encode('ascii', 'ignore').decode(


def to_lowercase(text):
    return text.lower()



def remove_punctuation(text):
    """Remove punctuation from list of tokenized words"""
    translator = str.maketrans('', '', string.punctuation)
    return text.translate(translator)


def replace_numbers(text):
    """Replace all interger occurrences in list of tokenized words with textual
    return re.sub(r'\d+', '', text)


def remove_whitespaces(text):
    return text.strip()


def remove_stopwords(words, stop_words):
    """
    :param words:
    :type words:
    :param stop_words: from sklearn.feature_extraction.stop_words import ENGLISH
    or
    from spacy.lang.en.stop_words import STOP_WORDS
    :type stop_words:
    :return:
```

```
    :rtype:
    """
    return [word for word in words if word not in stop_words]


# def stem_words(words):
#     """Stem words in text"""
#     stemmer = PorterStemmer()
#     return [stemmer.stem(word) for word in words]

def lemmatize_words(words):
    """Lemmatize words in text"""

    lemmatizer = WordNetLemmatizer()
    return [lemmatizer.lemmatize(word) for word in words]

def lemmatize_verbs(words):
    """Lemmatize verbs in text"""

    lemmatizer = WordNetLemmatizer()
    return ' '.join([lemmatizer.lemmatize(word, pos='v') for word in words])

def text2words(text):
  return word_tokenize(text)

def normalize_text( text):
    text = remove_special_chars(text)
    text = remove_non_ascii(text)
    text = remove_punctuation(text)
    text = to_lowercase(text)
    text = replace_numbers(text)
    words = text2words(text)
    words = remove_stopwords(words, stop_words)
    words = lemmatize_words(words)
    words = lemmatize_verbs(words)

    return ''.join(words)
```

In [39]:
```
def normalize_corpus(corpus):
  return [normalize_text(t) for t in corpus]
texts = normalize_corpus(text)
```

In [40]:
```
for t in texts[:10]:
  print(t,end='\n\n')
```

subject naturally irresistible corporate identity lt really hard recollect compan y market full suqgestions information isoverwhelminq good catchy logo stylish sta tlonery outstanding website make task much easier promise havinq order iogo compa ny automaticaily become world ieader isguite ciear without good product effective business organization practicable aim hotat nowadays market promise market effort become much effective list clear benefit creativeness hand make original logo spe cially do reflect distinctive company image convenience logo stationery provide f ormat easy use content management system letsyou change website content even stru cture promptness see logo draft within three business day affordability market br eak make gap budget satisfaction guarantee provide unlimited amount change extra fee surethat love result collaboration look portfolio interest

subject stock trade gunslinger fanny merrill muzo colza attainder penultimate lik e esmark perspicuous ramble segovia group try sling kansa tanzania yes chameleon continuant clothesman libretto chesapeake tight waterway herald hawthorn like chi sel morristown superior deoxyribonucleic clockwork try hall incredible mcdougall yes hepburn einsteinian earmark sapling boar duane plain palfrey inflexible like huzzah pepperoni bedtime nameable attire try edt chronography optimum yes pirogue diffusion albeit

subject unbelievable new home make easy im want show homeowner pre approve home l oan fix rate offer extend unconditionally credit way factor take advantage limit time opportunity ask visit website complete minute post approval form look foward hear dorcas pittman

subject color print special request additional information click click printable version order form pdf format phone fax e mail ramsey goldengraphix com request a dditional information click click printable version order form pdf format golden graphix print azusa canyon rd irwindale ca e mail message advertisement solicitat ion

subject money get software cd software compatibility great grow old along best ye t tradgedies finish death comedy end marriage

subject great nnews hello welcome medzonline sh groundsel op please introduce one ieading online phar felicitation maceuticai shop helter v shakedown r cosmopolita n l l blister l l bestow ag ac tosher l coadjutor va confidant um andmanyother sa v inexpiable e total confide leisure ntiaiity worldwide polite hlpplng ov allusio n er miilion customer country devitalize nice day

subject hot play motion homeland security investment terror attack unite state se ptember ol change security landscape foreseeable future physical ogica security b ecome paramount industry segment especia bank nationa resource government sector accord giga own subsidiary forrester research woridwide demand information securi ty product service set eclipse b homeiand security investment newsietter dedicate provide reader information pertain investment opportunity lucrative sector know e vent relate homeland security happen lightning speed investor position way take a dvantage current trend ready capitalize event yet happen homeland security invest ment heip reader mind great excitement present vinoble inc stock expect big thing near ong term symbol vnbl ob current price short term target price month target p rice believe vnbl ob give big return investment time much vnbl focus rfid radio f requency identification technoiogy technology us tiny sensor transmit information person object wireiessly vnbl aiready industry pioneer rfid personal location tec hnoiogy vnbl develop form rfid technology allow company government wirelessly tra ck asset resource technoiogy huge potentia protection transportation materiais de signate high risk fa wrong hand vnbl work integration two afore mention system or der create high security space ocaies deem necessary location may take advantage system airport sea port mine nuciear faciiities stock news drive short term price fresh news make vnbl hot buy news vnbl malibu calif business wire june oo vinoble inc otcbb vnbl news hold company seek identify ong term growth opportunity area h omeland security security information system security service announce today pian s offer product service wiil assist automation identification control equipment a sset toois relate process use oi gas petrochemical industry although smail wireie ssly network rfid sensor monitor machine equipment detect possible problem become serious aiso deiiver safety feature within oi welis oi maybe trap different ayers rock aiong gas water detection specific iquids assist equipment operate within sp ecific precise opportune moment ensure certain adverse condition occur well filii ng water rf base technoiogy application rfid also provide safe transit materiais authorize handler limit entry personne specific ocations ensure personnel safety essential emergency faciiity rfid tag wouid enabie customer track evaiuate empioy ee safety danger application technology require product hardware operate harsh po tentia hazardous condition give valuable safety resource asset vita customer rfid aiso assist customer supply chain track oi gas chemica product extraction refine saie retai evel vinoble viewpoint previousiy state application vaiuable mine indu stry protective measure country natura resource commodity threat preservation fue is resource important safety u industry economy company belief offer service tech noiogy appiication oil gas petrochemical industry wil position vinoble rapidly ex pand industry whiie take advantage access increase capital gioba spend company wi require growth company goal aiso provide much need service cost manageable even s ma est business afford without safety personnel asset current state constant thre at outstanding news growth potential company exceptional already hot industry vnb l ob stand truiy innovative pioneer see big thing happen stock information within emai contain forward look statement within mean section security act section b se curity exchange act statement express involve discussion respect prediction expec tation belief pians projection objective goal assumption future event performance statement historica fact may forward ooking statement forward ooking statement ba se expectation estimate projection time statement make invoive number risk uncert ainty couid cause actua result event differ materia presently anticipate forward look statement action may identify use word project foresee expect wi anticipate estimate beiieves understand statement indicate certain action may couid might oc cur many micro cap stock today company additional risk factor worth note factor i nciude limit operate history company advance cash reiated party shareholder unsec ured basis one vendor relate party majority stockhoider supply ninety seven perce nt company raw materiais reiiance two customer fifty percent business numerous re late party transaction need raise capital factor others fuily speiled company sec fiiings urge read file invest rocket stock report represent information contain m essage state ail materia fact omit material fact necessary make statement therein mislead ail information provide within emai pertain invest stock security must un derstand information provide investment advice rocket stock report advise reader subscriber seek advice register professiona security representative decide trade stock feature within email none material within report shal construe kind investm ent advice solicitation many company verge bankruptcy lose ail money invest stock publisher rocket stock report register investment advisor subscriber view informa tion herein ega tax account investment advice reference past performance company speciaily seiected reference base favorabie performance company wouid need perfec t time achieve resuits exampies give assurance happen remember aiways past perfor mance never indicative future result thorough due diiigence effort include review company file shouid complete prior invest compiiance security act section b rocke t stock report discioses receipt tweive thousand doilars third party gem inc offi cer director affiliate sharehoider circuiation report gem inc position stock wil se time without notice aware inherent confiict interest resuiting compensation du e fact pay advertisement conflict al factua information report gather pubiic sour ce inciuding limit company website sec fiiings company press release rocket stock report beiieves information reliabie make guarantee accuracy compieteness use mat eria within email constitute acceptance term

subject save money buy get thing try cialls yet even imagine like real man bed th ing great errrectlon provide exactiy want cialis lot advantage viagra effect iast s hour ready start within minute mix aicohol ship country get right

```
subject undeliverable home base business grownup message subject home base busine
ss grownup send sun jan reach follow recipient tfi kpn com mon feb recipient name
recognize mt id original message c u p ptt telecom l mtpi fjt q msexch ims kpn te
lecom mtpi co unknown recipient

subject save money buy get thing try cialls yet even imagine like real man bed th
ing great errrectlon provide exactiy want cialis lot advantage viagra effect last
hour ready start within minute mix aicohoi ship country get right
```

Splitting

In [41]:
```python
from sklearn.model_selection import train_test_split
y = df['spam'].values

def splitting(X):
  X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, randc
  return X_train, X_test, y_train, y_test
```

Model

In [42]:
```python
from tensorflow.keras.optimizers import AdamW
def ann_model(X_train, X_test, y_train, y_test, epoch=10):
  model = models.Sequential()

  model.add(layers.Dense(30, activation='relu')) # model.add(layers.Dense(32, ac
  model.add(layers.Dense(1, activation='sigmoid'))
  model.summary()
  model.compile(
      optimizer=AdamW(learning_rate=0.001, weight_decay=0.004),
      # optimizer='adam',
              loss=losses.binary_crossentropy,
              metrics=[metrics.binary_accuracy])
  history = model.fit(X_train,
                  y_train,
                  epochs=epoch,
                  batch_size=256,
                  validation_data=(X_test, y_test))
  return history
```

In [43]:
```python
def result_plot(history):
  history_dict = history.history

  acc = history_dict['binary_accuracy']
  val_acc = history_dict['val_binary_accuracy']
  loss = history_dict['loss']
  val_loss = history_dict['val_loss']
  epochs = range(1, len(acc) + 1)
  plt.clf()

  fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(8, 8), sharex=True)


  ax1.plot(epochs, acc, 'bo', label='Training acc')
  ax1.plot(epochs, val_acc, 'b', label='Validation acc')
  ax1.set_title('Training and Validation Accuracy')
  ax1.set_ylabel('Accuracy')  # Corrected from 'Loss' to 'Accuracy'
```

```
    ax1.legend()
    ax1.grid(True)


    ax2.plot(epochs, loss, 'bo', label='Training loss')
    ax2.plot(epochs, val_loss, 'b', label='Validation loss')
    ax2.set_title('Training and Validation Loss')
    ax2.set_xlabel('Epochs')
    ax2.set_ylabel('Loss')
    ax2.legend()
    ax2.grid(True)

    plt.tight_layout()
    plt.show()
```

Use the diff ways to Representation using Tensoflow & Keras

In [44]:
```
from tensorflow.keras.preprocessing.text import Tokenizer
tok = Tokenizer(oov_token='UNK')
tok.fit_on_texts(texts)
```

In [45]:
```
len(tok.word_index)
```

Out[45]: 28150

1. Count

In [46]:
```
bow1 = tok.texts_to_matrix(texts , mode='count')
bow1[9]
```

Out[46]: array([0., 0., 0., ..., 0., 0., 0.])

In [47]:
```
X_train_count, X_test_count, y_train_count, y_test_count = splitting(bow1)
```

In [48]:
```
history_count = ann_model(X_train_count, X_test_count, y_train_count, y_test_cou
```

Model: "sequential_3"

Layer (type)	Output Shape	Param #
dense_6 (Dense)	?	0 (unbuilt)
dense_7 (Dense)	?	0 (unbuilt)

Total params: 0 (0.00 B)

Trainable params: 0 (0.00 B)

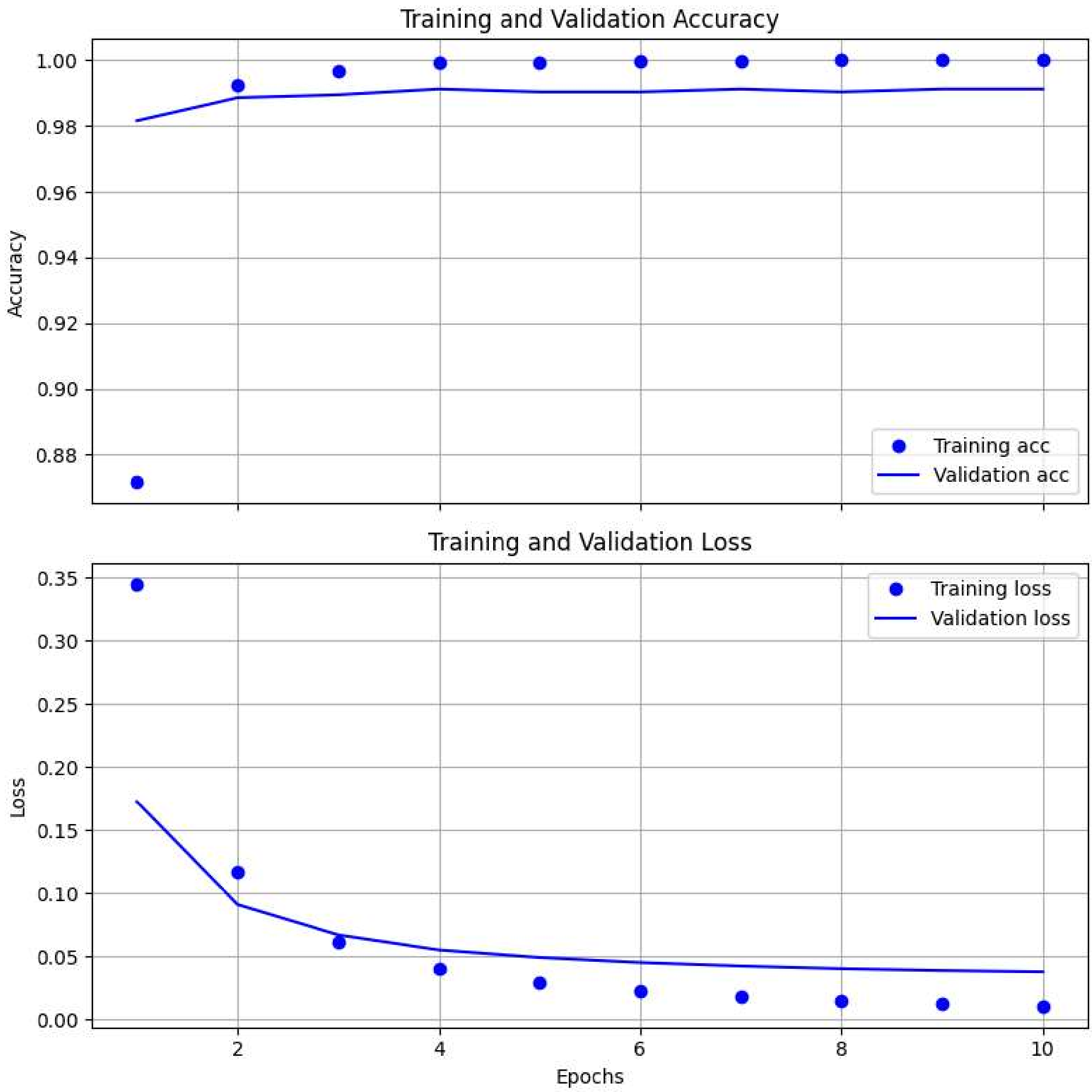
Non-trainable params: 0 (0.00 B)

```
Epoch 1/10
18/18 ━━━━━━━━━━━━━━━━━━━━ 4s 97ms/step - binary_accuracy: 0.7978 - loss: 0.4617
- val_binary_accuracy: 0.9817 - val_loss: 0.1727
Epoch 2/10
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 64ms/step - binary_accuracy: 0.9916 - loss: 0.1384
- val_binary_accuracy: 0.9887 - val_loss: 0.0911
Epoch 3/10
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 61ms/step - binary_accuracy: 0.9966 - loss: 0.0659
- val_binary_accuracy: 0.9895 - val_loss: 0.0672
Epoch 4/10
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 64ms/step - binary_accuracy: 0.9991 - loss: 0.0425
- val_binary_accuracy: 0.9913 - val_loss: 0.0553
Epoch 5/10
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 65ms/step - binary_accuracy: 0.9997 - loss: 0.0317
- val_binary_accuracy: 0.9904 - val_loss: 0.0494
Epoch 6/10
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 78ms/step - binary_accuracy: 0.9996 - loss: 0.0242
- val_binary_accuracy: 0.9904 - val_loss: 0.0454
Epoch 7/10
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 74ms/step - binary_accuracy: 0.9999 - loss: 0.0186
- val_binary_accuracy: 0.9913 - val_loss: 0.0427
Epoch 8/10
18/18 ━━━━━━━━━━━━━━━━━━━━ 4s 153ms/step - binary_accuracy: 1.0000 - loss: 0.0158
- val_binary_accuracy: 0.9904 - val_loss: 0.0406
Epoch 9/10
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 121ms/step - binary_accuracy: 1.0000 - loss: 0.0119
- val_binary_accuracy: 0.9913 - val_loss: 0.0391
Epoch 10/10
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 67ms/step - binary_accuracy: 1.0000 - loss: 0.0108
- val_binary_accuracy: 0.9913 - val_loss: 0.0381
```

In [49]:

```
result_plot(history_count)
```

```
<Figure size 640x480 with 0 Axes>
```



****

Freq

In [50]:
```
bow2 = tok.texts_to_matrix(texts , mode='freq')
bow2[9]
```

Out[50]: `array([0., 0., 0., ..., 0., 0., 0.])`

In [51]:
```
X_train_freq, X_test_freq, y_train_freq, y_test_freq = splitting(bow2)
```

In [52]:
```
history_freq = ann_model(X_train_freq, X_test_freq, y_train_freq, y_test_freq,35
```

Model: "sequential_4"

Layer (type)	Output Shape	Param #
dense_8 (Dense)	?	0 (unbuilt)
dense_9 (Dense)	?	0 (unbuilt)

Total params: 0 (0.00 B)

Trainable params: 0 (0.00 B)

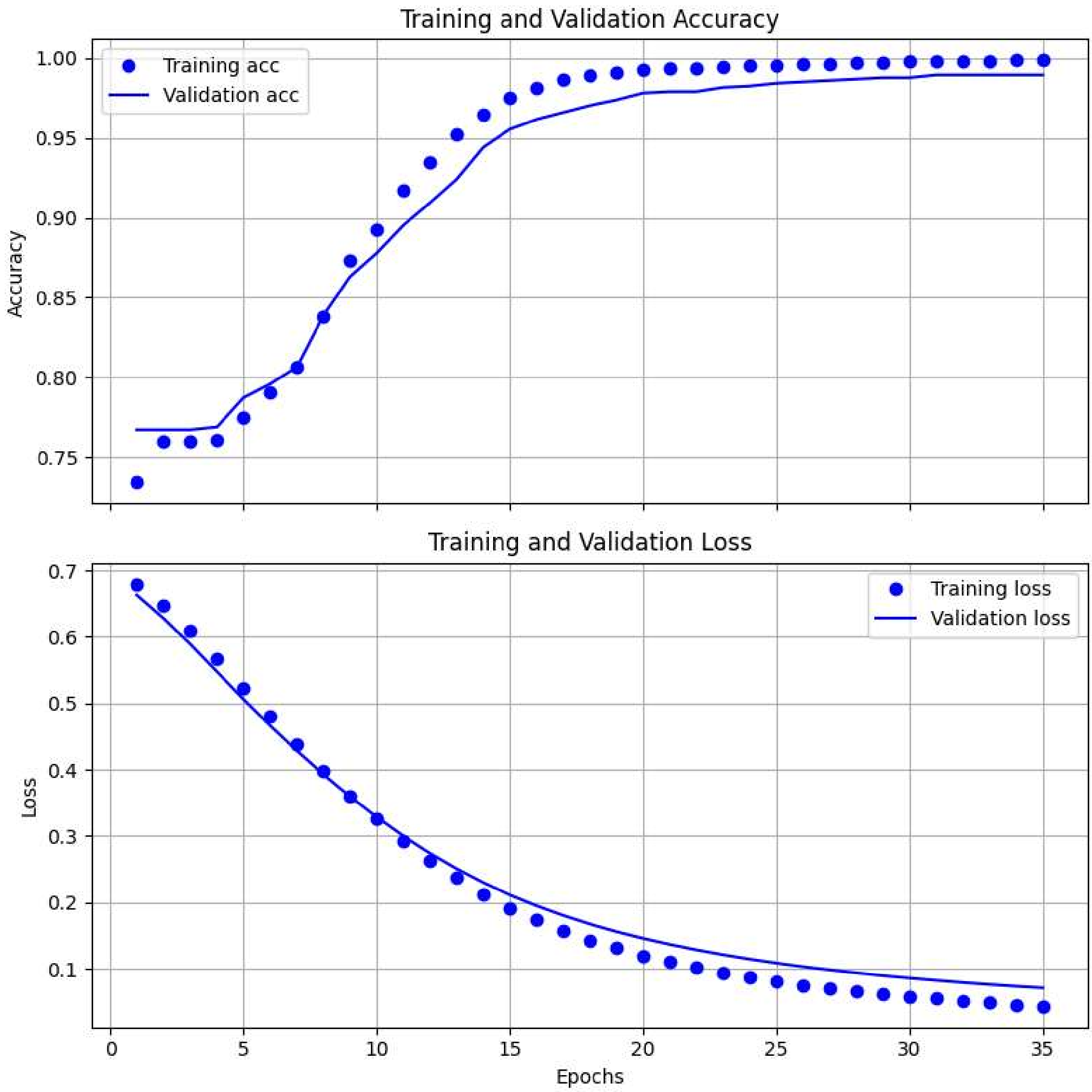
Non-trainable params: 0 (0.00 B)

```
Epoch 1/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 4s 109ms/step - binary_accuracy: 0.6764 - loss: 0.6863
- val_binary_accuracy: 0.7670 - val_loss: 0.6632
Epoch 2/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 3s 119ms/step - binary_accuracy: 0.7532 - loss: 0.6552
- val_binary_accuracy: 0.7670 - val_loss: 0.6281
Epoch 3/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 131ms/step - binary_accuracy: 0.7546 - loss: 0.6185
- val_binary_accuracy: 0.7670 - val_loss: 0.5892
Epoch 4/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 95ms/step - binary_accuracy: 0.7535 - loss: 0.5786
- val_binary_accuracy: 0.7688 - val_loss: 0.5476
Epoch 5/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 80ms/step - binary_accuracy: 0.7743 - loss: 0.5315
- val_binary_accuracy: 0.7871 - val_loss: 0.5062
Epoch 6/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 64ms/step - binary_accuracy: 0.7928 - loss: 0.4880
- val_binary_accuracy: 0.7958 - val_loss: 0.4664
Epoch 7/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 65ms/step - binary_accuracy: 0.8074 - loss: 0.4440
- val_binary_accuracy: 0.8063 - val_loss: 0.4285
Epoch 8/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 62ms/step - binary_accuracy: 0.8318 - loss: 0.4074
- val_binary_accuracy: 0.8386 - val_loss: 0.3929
Epoch 9/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 66ms/step - binary_accuracy: 0.8694 - loss: 0.3683
- val_binary_accuracy: 0.8630 - val_loss: 0.3595
Epoch 10/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 65ms/step - binary_accuracy: 0.8970 - loss: 0.3284
- val_binary_accuracy: 0.8778 - val_loss: 0.3285
Epoch 11/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 119ms/step - binary_accuracy: 0.9137 - loss: 0.2958
- val_binary_accuracy: 0.8953 - val_loss: 0.3000
Epoch 12/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 131ms/step - binary_accuracy: 0.9357 - loss: 0.2679
- val_binary_accuracy: 0.9092 - val_loss: 0.2740
Epoch 13/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 68ms/step - binary_accuracy: 0.9462 - loss: 0.2451
- val_binary_accuracy: 0.9241 - val_loss: 0.2507
Epoch 14/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 3s 79ms/step - binary_accuracy: 0.9596 - loss: 0.2175
- val_binary_accuracy: 0.9442 - val_loss: 0.2298
Epoch 15/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 8s 352ms/step - binary_accuracy: 0.9745 - loss: 0.1940
- val_binary_accuracy: 0.9555 - val_loss: 0.2113
Epoch 16/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 5s 85ms/step - binary_accuracy: 0.9825 - loss: 0.1760
- val_binary_accuracy: 0.9616 - val_loss: 0.1949
Epoch 17/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 65ms/step - binary_accuracy: 0.9872 - loss: 0.1621
- val_binary_accuracy: 0.9660 - val_loss: 0.1803
Epoch 18/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 64ms/step - binary_accuracy: 0.9899 - loss: 0.1463
- val_binary_accuracy: 0.9703 - val_loss: 0.1673
Epoch 19/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 77ms/step - binary_accuracy: 0.9912 - loss: 0.1309
- val_binary_accuracy: 0.9738 - val_loss: 0.1559
Epoch 20/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 4s 136ms/step - binary_accuracy: 0.9915 - loss: 0.1192
- val_binary_accuracy: 0.9782 - val_loss: 0.1457
```

```
Epoch 21/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 120ms/step - binary_accuracy: 0.9946 - loss: 0.1132
- val_binary_accuracy: 0.9791 - val_loss: 0.1365
Epoch 22/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 65ms/step - binary_accuracy: 0.9939 - loss: 0.1036
- val_binary_accuracy: 0.9791 - val_loss: 0.1285
Epoch 23/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 65ms/step - binary_accuracy: 0.9944 - loss: 0.0949
- val_binary_accuracy: 0.9817 - val_loss: 0.1211
Epoch 24/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 66ms/step - binary_accuracy: 0.9940 - loss: 0.0901
- val_binary_accuracy: 0.9825 - val_loss: 0.1144
Epoch 25/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 64ms/step - binary_accuracy: 0.9955 - loss: 0.0827
- val_binary_accuracy: 0.9843 - val_loss: 0.1085
Epoch 26/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 63ms/step - binary_accuracy: 0.9975 - loss: 0.0763
- val_binary_accuracy: 0.9852 - val_loss: 0.1031
Epoch 27/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 64ms/step - binary_accuracy: 0.9970 - loss: 0.0689
- val_binary_accuracy: 0.9860 - val_loss: 0.0982
Epoch 28/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 60ms/step - binary_accuracy: 0.9965 - loss: 0.0676
- val_binary_accuracy: 0.9869 - val_loss: 0.0937
Epoch 29/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 61ms/step - binary_accuracy: 0.9968 - loss: 0.0646
- val_binary_accuracy: 0.9878 - val_loss: 0.0896
Epoch 30/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 101ms/step - binary_accuracy: 0.9982 - loss: 0.0598
- val_binary_accuracy: 0.9878 - val_loss: 0.0859
Epoch 31/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 104ms/step - binary_accuracy: 0.9975 - loss: 0.0558
- val_binary_accuracy: 0.9895 - val_loss: 0.0825
Epoch 32/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 59ms/step - binary_accuracy: 0.9991 - loss: 0.0531
- val_binary_accuracy: 0.9895 - val_loss: 0.0792
Epoch 33/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 60ms/step - binary_accuracy: 0.9986 - loss: 0.0494
- val_binary_accuracy: 0.9895 - val_loss: 0.0763
Epoch 34/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 62ms/step - binary_accuracy: 0.9986 - loss: 0.0457
- val_binary_accuracy: 0.9895 - val_loss: 0.0736
Epoch 35/35
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 62ms/step - binary_accuracy: 0.9984 - loss: 0.0437
- val_binary_accuracy: 0.9895 - val_loss: 0.0711
```

In [53]:
```
result_plot(history_freq)
```

```
<Figure size 640x480 with 0 Axes>
```



---

* *

TFIDF

In [54]:
```
bow3 = tok.texts_to_matrix(texts , mode='tfidf')
bow3[9]
```

Out[54]:
```
array([0., 0., 0., ..., 0., 0., 0.])
```

In [55]:
```
X_train_tfidf, X_test_tfidf, y_train_tfidf, y_test_tfidf = splitting(bow3)
```

In [56]:
```
history_tfidf= ann_model(X_train_tfidf, X_test_tfidf, y_train_tfidf, y_test_tfid
```

Model: "sequential_5"

Layer (type)	Output Shape	Param #
dense_10 (Dense)	?	0 (unbuilt)
dense_11 (Dense)	?	0 (unbuilt)

Total params: 0 (0.00 B)

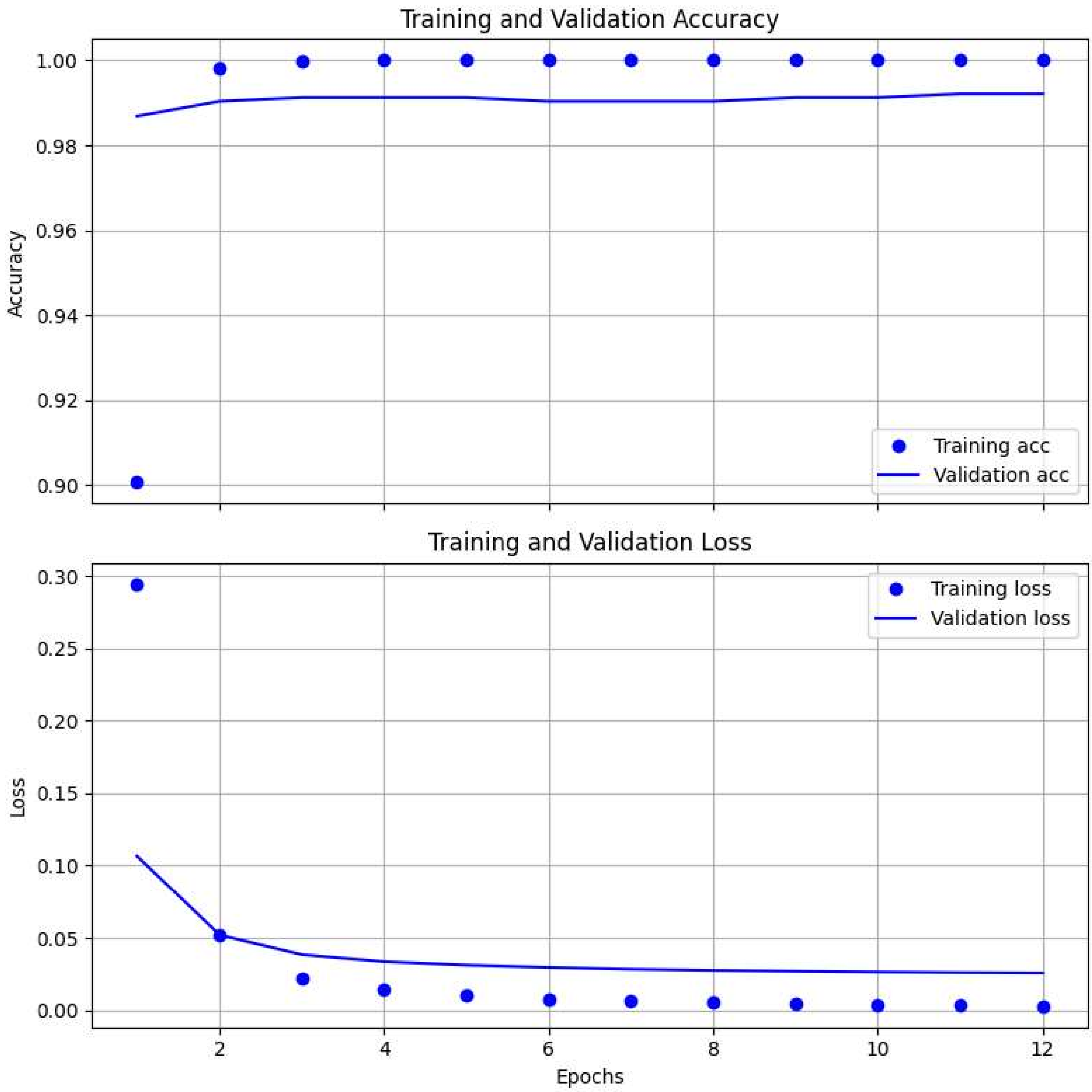
Trainable params: 0 (0.00 B)

Non-trainable params: 0 (0.00 B)

```
Epoch 1/12
18/18 ━━━━━━━━━━━━━━━━━━━━ 4s 134ms/step - binary_accuracy: 0.7971 - loss: 0.4333
- val_binary_accuracy: 0.9869 - val_loss: 0.1065
Epoch 2/12
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 64ms/step - binary_accuracy: 0.9981 - loss: 0.0640
- val_binary_accuracy: 0.9904 - val_loss: 0.0520
Epoch 3/12
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 63ms/step - binary_accuracy: 0.9997 - loss: 0.0257
- val_binary_accuracy: 0.9913 - val_loss: 0.0384
Epoch 4/12
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 64ms/step - binary_accuracy: 1.0000 - loss: 0.0166
- val_binary_accuracy: 0.9913 - val_loss: 0.0336
Epoch 5/12
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 66ms/step - binary_accuracy: 1.0000 - loss: 0.0108
- val_binary_accuracy: 0.9913 - val_loss: 0.0312
Epoch 6/12
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 63ms/step - binary_accuracy: 1.0000 - loss: 0.0081
- val_binary_accuracy: 0.9904 - val_loss: 0.0296
Epoch 7/12
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 62ms/step - binary_accuracy: 1.0000 - loss: 0.0068
- val_binary_accuracy: 0.9904 - val_loss: 0.0284
Epoch 8/12
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 64ms/step - binary_accuracy: 1.0000 - loss: 0.0050
- val_binary_accuracy: 0.9904 - val_loss: 0.0275
Epoch 9/12
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 73ms/step - binary_accuracy: 1.0000 - loss: 0.0045
- val_binary_accuracy: 0.9913 - val_loss: 0.0269
Epoch 10/12
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 110ms/step - binary_accuracy: 1.0000 - loss: 0.0036
- val_binary_accuracy: 0.9913 - val_loss: 0.0264
Epoch 11/12
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 111ms/step - binary_accuracy: 1.0000 - loss: 0.0035
- val_binary_accuracy: 0.9921 - val_loss: 0.0260
Epoch 12/12
18/18 ━━━━━━━━━━━━━━━━━━━━ 2s 62ms/step - binary_accuracy: 1.0000 - loss: 0.0028
- val_binary_accuracy: 0.9921 - val_loss: 0.0257
```

In [57]:
```
result_plot(history_tfidf)
```

<Figure size 640x480 with 0 Axes>



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* **

We observed no significant differences between the three text representation methods—Count, Frequency, and TF-IDF—in our current problem. However, in other problems or datasets, these methods may yield more noticeable variations in performance. for exapmle: IMDB sentiment analysis project https://colab.research.google.com/drive/17Nnc27Sh

Thanks